SMART ROADMAP BUILDER

Software Requirements Specification

AI-Powered Personalized Learning Platform

Version 1.0 · 2025

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1. INTRODUCTION

This document defines the complete set of Functional and Non-Functional Requirements for Smart Roadmap Builder an AI-powered web application that generates personalized learning roadmaps and helps users stay consistent through intelligent tracking, contextual reminders, and adaptive scheduling.

The platform targets self-learners, career-switchers, and upskilling professionals who need a structured, adaptive, and motivating learning environment.

2. FUNCTIONAL REQUIREMENTS

The following 33 requirements define what the Smart Roadmap Builder must do. They are organized by feature domain and written at a level of detail sufficient for engineering scoping and QA test case derivation.

ID	Category	Requirement
FR-01	**Authentication**	Users can register via email/password or OAuth providers (Google, GitHub). Account activation requires email verification before access is granted.
FR-02	**Authentication**	Users can log in using email/password or a connected OAuth provider. Failed login attempts are throttled after 5 consecutive failures.
FR-03	**Authentication**	Users can reset their password via a time-limited (60-minute) email link. Old password is invalidated immediately upon reset.
FR-04	**Authentication**	Users can enable two-factor authentication (2FA) via an authenticator app (TOTP). 2FA can be toggled from the Security Settings page.
FR-05	**Authentication**	Users can manage active sessions, view device/location info, and remotely log out any session from the Security Settings page.
FR-06	**Onboarding**	New users are guided through a structured onboarding wizard (3–5 steps) that captures their learning goal, current skill level, available weekly hours, and preferred learning style.
FR-07	**Onboarding**	The platform accepts a free-text goal input and maps it to a predefined taxonomy of skills and topics using AI-assisted classification.
FR-08	**Onboarding**	Users can import an existing resume or LinkedIn profile during onboarding to auto-populate their current skills and experience.
FR-09	**Onboarding**	Users can skip onboarding and manually configure their profile and roadmap preferences at any time from the Settings page.
FR-10	**Roadmap Generation**	The AI engine generates a structured learning roadmap composed of ordered milestones, each containing a set of tasks with estimated durations and resource recommendations.

ID	Category	Requirement
FR-11	Roadmap Generation	Each generated roadmap includes a start date, estimated completion date, and a weekly learning plan aligned to the user's declared availability.
FR-12	Roadmap Generation	Users can regenerate any roadmap section or adjust parameters (pace, depth, preferred content type) and receive an updated plan without losing prior progress.
FR-13	Roadmap Generation	The platform suggests curated external resources (articles, videos, courses, exercises) for each task, ranked by relevance and community rating.
FR-14	Roadmap Generation	Users can add, remove, reorder, or merge roadmap tasks and milestones using a drag-and-drop interface. Manual edits are preserved across AI regenerations.
FR-15	Task Tracking	Users can mark tasks as Not Started, In Progress, or Completed. Status changes are timestamped and reflected in real-time across all views.
FR-16	Task Tracking	A daily task queue surfaces the user's prioritized to-do list for the current day based on their roadmap schedule and outstanding items.
FR-17	Task Tracking	Users can log time spent on a task manually or via an in-app timer. Logged hours are used to refine future duration estimates.
FR-18	Task Tracking	Users can attach personal notes, bookmarks, and file links to any task for reference during and after completion.
FR-19	Progress & Analytics	A progress dashboard displays overall roadmap completion percentage, milestone status, weekly activity streak, and time invested per category.
FR-20	Progress & Analytics	The platform compares the user's actual pace against the planned schedule and surfaces a clear on-track / behind / ahead status indicator.
FR-21	Progress & Analytics	Users can view a historical activity log (calendar heatmap view) showing daily learning sessions over the past 12 months.
FR-22	Progress & Analytics	Users can export their progress report as a PDF or CSV, including milestones completed, time logged, and skills acquired.
FR-23	Notifications	Users can configure daily or weekly learning reminders via email, browser push notification, or in-app alert, including custom time preferences.
FR-24	Notifications	The platform sends automatic nudges when a user has been inactive for more than 3 consecutive days (threshold configurable by the user).
FR-25	Notifications	Users receive in-app and email notifications for milestone completions, streak achievements, and roadmap schedule changes.
FR-26	Notifications	All notification types can be individually enabled or disabled per channel (email, push, in-app) from the Notifications Settings page.
FR-27	Smart Rescheduling	When a user falls behind schedule, the AI automatically proposes a revised roadmap timeline that redistributes remaining tasks across available future slots.

ID	Category	Requirement
FR-28	**Smart Rescheduling**	Users can block out unavailable dates (e.g., vacations, deadlines) on the calendar. The scheduler excludes these dates from rescheduling calculations.
FR-29	**Smart Rescheduling**	The rescheduling engine respects task dependencies and preserves milestone ordering when redistributing tasks across the updated timeline.
FR-30	**Smart Rescheduling**	Users are prompted to accept or reject AI-proposed schedule changes before they are applied. A diff view highlights what would change.
FR-31	**Profile & Settings**	Users can update their display name, avatar, learning goal, weekly availability, and preferred content formats at any time from the Profile page.
FR-32	**Profile & Settings**	Users can delete their account and all associated data permanently. A 30-day grace period allows account recovery before final deletion.
FR-33	**Profile & Settings**	Users can maintain multiple independent roadmaps (e.g., one per learning goal) and switch between them from a unified roadmap switcher in the navigation.

3. NON-FUNCTIONAL REQUIREMENTS

The following requirements define the quality attributes and operational constraints that govern how the system performs. Each requirement includes a measurable target to enable objective acceptance testing.

Category	Requirement	Measurable Target
Performance	AI roadmap generation must complete end-to-end within a defined latency budget under normal load.	*≤ 5 seconds (p95) for initial roadmap generation with up to 30 tasks.*
Performance	All core UI interactions (page loads, task status updates, dashboard rendering) must feel instantaneous under normal conditions.	*≤ 1.5 seconds (p95) TTFB; ≤ 300ms for API read responses.*
Performance	The system must handle concurrent active users without performance degradation during peak usage hours.	*Sustain ≥ 5,000 concurrent users with < 5% increase in p95 latency.*
Scalability	The backend architecture must scale horizontally to accommodate user growth without requiring service redesign.	*Auto-scaling triggers at 70% CPU/memory; supports 10× traffic spikes within 2 minutes.*
Scalability	The database layer must handle growing user datasets and roadmap content without degrading query performance.	*Query response ≤ 200ms for datasets up to 10M user records using indexed queries.*
Availability	The platform must maintain high availability and minimize unplanned downtime to protect user learning continuity.	*≥ 99.9% uptime SLA (< 8.7 hours downtime/year); planned maintenance < 4 hours/month.*

Category	Requirement	Measurable Target
Availability	Scheduled maintenance windows must be communicated in advance and must not occur during peak usage hours.	*≥ 72 hours advance notice; maintenance windows between 02:00–04:00 UTC only.*
Security	All data transmitted between client and server must be encrypted in transit using modern TLS standards.	*TLS 1.2 minimum; TLS 1.3 preferred. Automatic HTTP → HTTPS redirect enforced.*
Security	All sensitive user data (passwords, tokens) must be stored securely at rest using industry-standard hashing and encryption.	*Passwords hashed with bcrypt (cost ≥ 12); PII encrypted with AES-256 at rest.*
Security	The platform must protect against common web vulnerabilities as classified by OWASP Top 10.	*Zero critical/high OWASP findings in quarterly penetration tests and automated scans.*
Security	Access to user data must be enforced through strict authentication and authorization controls with full audit logging.	*All admin data access logged with user ID, timestamp, and action. Logs retained ≥ 12 months.*
Usability	The platform must be accessible and usable on major modern browsers across desktop and mobile screen sizes.	*Responsive layout for screens ≥ 360px; tested on Chrome, Firefox, Safari, Edge (latest 2 versions).*
Usability	The UI must comply with WCAG 2.1 Level AA accessibility standards to ensure inclusivity for users with disabilities.	*100% compliance with WCAG 2.1 AA; verified by automated audit + manual screen reader testing.*
Usability	New users must be able to complete onboarding and generate their first roadmap within a short, frictionless session.	*Median time-to-first-roadmap ≤ 5 minutes, measured via session analytics.*
Reliability	The platform must implement data redundancy and automated backups to prevent data loss in the event of infrastructure failure.	*Daily automated backups; RPO ≤ 1 hour; RTO ≤ 4 hours. Backups retained for 30 days.*
Maintainability	The codebase must be structured to allow rapid feature iteration, independent service deployment, and straightforward debugging.	*Modular architecture (≥ 80% unit test coverage); MTTR for P1 incidents ≤ 2 hours.*
Data Privacy	The platform must comply with applicable data privacy regulations and provide users with full data transparency and control.	*GDPR-compliant; data export delivered within 72 hours; deletion completed within 30 days of request.*

4. GLOSSARY

Roadmap A structured, AI-generated plan composed of ordered milestones and tasks aligned to a specific learning goal.

Milestone A major checkpoint in a roadmap representing the completion of a thematic learning phase (e.g., 'Core Python Fundamentals').

Adaptive Scheduling The system's ability to automatically redistribute uncompleted tasks across future time slots when the user deviates from their original plan.

OAuth Open Authorization — an industry-standard protocol allowing users to authenticate using a third-party identity provider (e.g., Google, GitHub).

TOTP Time-Based One-Time Password — a 2FA method using a rotating 6-digit code generated by an authenticator app.

RPO Recovery Point Objective — the maximum acceptable data loss measured in time (e.g., RPO ≤ 1 hour means at most 1 hour of data may be lost).

RTO Recovery Time Objective — the maximum acceptable time to restore service after a failure.

WCAG 2.1 AA Web Content Accessibility Guidelines version 2.1, Level AA — the standard for making digital content accessible to users with disabilities.